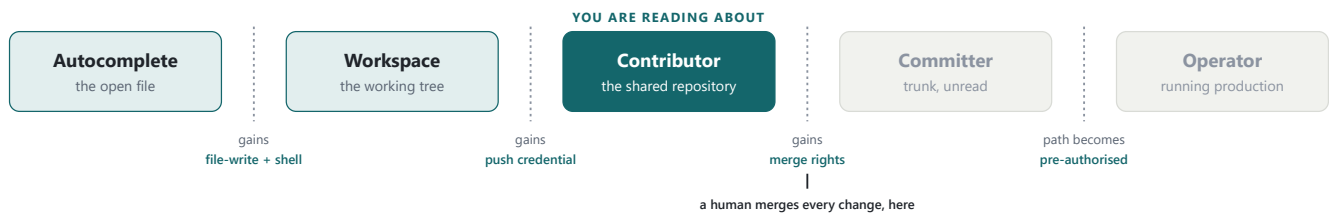


# Contributor

The agent gains a push credential: its output enters the shared repository as branches and pull requests, and a human merges every change. The large middle of observed practice — and the archetype the platform itself enforces.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Sources last verified 2026-09-13

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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## 01 What this archetype is

A Contributor ceiling permits an agent's output to reach **the shared repository with no human action in between**: the agent pushes branches and opens pull requests under its own identity, and a human merges every change. The name keeps the open-source governance intuition deliberately — a contributor opens pull requests that someone else merges — and the posture it encodes is that an agent's work should come through the same front door as anyone else's [ADR-0004](#).

This is the large middle of observed practice: the least-covered region of the spectrum because it makes a bad headline, and the one where nearly every named organisation delegating at scale actually sits. It is also the archetype the dominant platform pins teams into by construction: GitHub's own cloud agent *"cannot approve or merge a pull request"*, and its pull requests require an additional human approval [GitHub Docs, Aug 2026](#). A team that adopts a cloud or background agent lands here by default and stays here until an administrator deliberately builds a way out.

## 02 The boundary crossed

What separates Contributor from Workspace is a **push credential**: crossing this boundary, the agent's work leaves the machine and becomes shared state — visible to the team, running in CI, attributed to the agent's own identity — with no human relaying it. The boundary is a credential, enumerable and binary, which is what makes it a discontinuity rather than a matter of degree [● ADR-0004](#).

Two structural facts define the position. First, this is the archetype the tool market *ships*: background and cloud agents are products whose entire output surface is the pull request. Second, the boundary above is the only one on the spectrum enforced by the platform rather than by the team: **no product merges its own pull request on its own judgement** — the far-end survey found every real mechanism beyond this point has the shape of a human pre-authorising a merge, never an agent deciding one

[● Tooling Landscape, Aug 2026](#). Workspace is exited by adopting a tool; Contributor is exited only by an administrator changing a policy. That makes it the cheapest archetype to hold and the one many teams occupy without ever having decided to.

## 03 What a human still gates

The merge — every change. And on the best-run teams, two upstream gates as well: **invitation** (in Microsoft's `dotnet/runtime`, every agent PR *"was created at the explicit request of a human with maintainer rights... CCA cannot open PRs on its own"* [● Microsoft, Mar 2026](#)) and **task selection** (Monzo chooses per task what the agent may see; Atlassian runs a classifier tuned to 97% precision on *out-of-scope* work)

[● Practitioner Exemplars, Aug 2026](#).

What the merge gate is *worth* is this page's hardest question, because rubber-stamping under agent volume is now measured. A within-reviewer longitudinal study of 400 repeat reviewers across 11,429 reviews found approval rates on agent PRs rising **30.1% → 36.8%**, inline comments falling **−22%**, and review latency rising **+3.5×** — while the same reviewers' approval of human-authored PRs *declined*, ruling out general drift. Reviewers wait longer, look less, and approve more [● Habituation study, Jun 2026](#). In open source the erosion is further along: most AI-generated PRs receive no reviews at all, and where review occurs it is automation-mediated [● Duma et al., May 2026](#).

### INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. "A human merges every change" states who acts — whether the merge is a review or a relay is a separate, measurable question, and §06 treats protecting the difference as this archetype's engineering problem.

## 04 Who works this way

The anchor is **Stripe**. *"Over 1,300 Stripe pull requests... merged each week are completely minion-produced, and while they're human-reviewed, they contain no human-written code."* The containment is environmental, stated as the reason permission prompts can be skipped: devboxes are *"isolated from production resources*

and the internet", so minions "don't have access to real user data, Stripe's production services, or arbitrary network egress" — and the agent gets exactly two attempts at CI before the branch goes back to its human operator [● Stripe, Feb 2026](#).

The middle is well populated with named practice. **Monzo**, a regulated UK bank: "Agent Chip" authors ~10% of all merged PRs, runs 1,800+ tasks a day, and was built in-house for an explicitly stated withholding reason — *"We need to know exactly what the agent can see, what it can do, and constrain it as appropriate"*

[● Monzo, Aug 2026](#). **Dropbox**: "Nova" produces roughly 1 in 12 pull requests, a human makes the final judgment, and the four-stage measurement model (Fuel → Adoption → Output → Impact) is the most disciplined found — with no values published, an exemplary framework and a non-existent result

[● Dropbox, May 2026](#). **Uber**: 11% of PRs opened by agents, a purpose-built review-squeeze toolchain (Code Inbox, uReview, Autocover), and the corpus's most falsifiable cost admission — AI-related costs up 6× since 2024 [● Pragmatic Engineer, Mar 2026](#).

The best-measured public dataset at this archetype — in the whole corpus — is **Microsoft's**

**dotnet/runtime** : 878 agent PRs over ten months, **67.9% merged**, with a revert rate of **0.6% against 0.8%** for non-agent PRs, all in a public repository where the numbers can be independently checked

[● Microsoft, Mar 2026](#). One structural fact recurs across every account in this section, reported independently by almost all of them: **the constraint moved to review, and nobody reports having solved it**

[● Practitioner Exemplars, Aug 2026](#).

The archetype also carries a vendor's endorsement — to be read as a recommendation, dated, not as observed practice. Anthropic's *AI-Native SDLC Playbook* (2026) makes Contributor its recommended end state at the repository: *"Anything the agent writes arrives as a PR... the agent has no route to push to main"*, code-owner approval stays on every pull request, and the agent's reach is stated to stop one boundary short of production — *"the agent may act up to the production gate and cannot pass it."* Nowhere does the vendor recommend granting merge rights on the agent's judgement. The course publishes no measurements, so what it corroborates is this page's structural claim, not its numbers [● Anthropic playbook, L10 & L12, 2026](#).

## 05 What it gains you, what it costs

### WHAT IT GAINS YOU

- Authoring at volume with review preserved: unattended agents produce finished, CI-tested work while the team's existing merge discipline stays exactly where it was [● Stripe, Feb 2026](#).
- The same front door: agent work arrives attributed to the agent's own identity, through the same PR machinery, CI gates, and audit trail as everyone else's — the cleanest provenance of any archetype (§07).
- The cheapest ceiling to hold: the boundary above is platform-enforced, so staying here requires no engineering — only *not* building a bypass [● Tooling Landscape, Aug 2026](#).

### WHAT IT COSTS YOU

- The review queue becomes the bottleneck, with numbers: agentic PRs wait **5.3× longer** for reviewer pickup (1,055 vs 201 minutes), AI-assisted PRs run ~154% larger and are accepted at 32.7% against 84.5% unassisted [● LinearB, May 2026](#). The cost lands on colleagues: one maintainer opened nine agent PRs in a few hours and priced them at *"5 to 9 hours of review work, spread across team members... The bottleneck has moved"* [● Microsoft, Mar 2026](#).
- The load erodes the very control the archetype trusts: habituation is measured, and it worsens with the volume this archetype generates (§03).
- The platform bill scales with the PRs, not the merges: Uber's 6× cost growth is the only figure of its kind published, and it cuts against the productivity narrative [● Pragmatic Engineer, Mar 2026](#).

## 06 What it demands of verification

Nothing human was removed at this boundary — the human stopped *authoring*, not checking — so the entailment takes an unusual form here: what must be engineered is **the protection of the review itself**, against the volume the archetype generates. The habituation numbers in §03 are the forcing function: left unprotected, the merge gate degrades toward a relay, and the team drifts further along the spectrum whether or not it chose to [● Habituation study, Jun 2026](#).

The tools most teams name in their verification story cannot carry it. **The major vendors deliberately built their AI code reviewers so they cannot be the gate:** GitHub Copilot code review always leaves a non-counting *Comment* review, and Anthropic's check run *"always completes with a neutral conclusion"* — which GitHub's required-status-check rule counts as passing, so marking it required accomplishes nothing. The only independent evaluation of any AI reviewer found 56.3% of its comments rejected, and six of eight surveyed tools have no published accuracy figure from anyone, vendor included [● Verification: Gates, Aug 2026](#).

The vendor says as much in its own teaching. Anthropic's playbook states that its reviewer's *"Findings do not approve or block a PR on their own"*, that a platform engineer who wants to gate on them must build the gate from the check run's *"machine-readable tally"*, and that *"the agent that wrote the code has no way to*

*approve it.*" It names this archetype's throughput limit as reviewer attention — *"add sessions only while review is keeping up"* — and prescribes for evals what this section asks of every gate: live suites whose *"cases that once discriminated stop doing so"*, gated on pass rate in CI. One caution travels with the citation. The course's opening table promises *"human review reserved for regulated and critical code"*, a Committer-shaped claim the plays themselves never make — every play keeps code-owner approval on every pull request. Cite the plays, not the table [● Anthropic playbook, L1, L7, L9 & L10, 2026](#).

What the well-run occupants build instead: **gates that live server-side, outside the agent's reach** — CI and branch protection, not the agent harness (GitHub describes its own agent-side approval queue as *"a workflow convenience, not a security control"*, and Anthropic documents that its strongest agent-side stop is overridden by the harness after eight consecutive blocks) [● Verification: Gates, Aug 2026](#) — plus upstream demand control: invitation-only PR creation, task-eligibility classifiers, deterministic pipeline nodes for the steps that need no model, and hard retry caps that return failing work to a human rather than burning reviewer attention on the agent's third attempt [● Stripe, Feb 2026](#) [● Practitioner Exemplars, Aug 2026](#). The pattern across all of it: protect the reviewer's attention as an engineered budget, not an assumed resource.

## 07 Governance implications

**Provenance is at its cleanest here — and only here.** The agent authors under its own platform identity, so "which lines did a human write" has a recorded answer at the moment of contribution; merging remains adoption, not authorship — *"providing instructions to a machine and selecting an output does not equate to authorship"* [● Governance, Legal & IP, Aug 2026](#). The caveat travels with the record: the surviving attribution is platform-hosted, not repository-hosted, and dies in plain git under squash — the clean answer exists on the forge, not in every clone [● Governance, Legal & IP, Aug 2026](#).

One vendor artifact proposes a repository-hosted answer to a neighbouring question. Anthropic's playbook recommends committing the whole chain — an intent file, a specification, a plan, the diff and its tests, the pull request with its findings, the incident record — so that *"The chain of commits is also the audit trail: who asked for what, what the agent produced, and who approved it."* It answers "who asked for it" where the corpus's question was "which lines did a human write", and it is a recommended practice with no adoption evidence; but it lives in committed files, which survive the squash that kills co-author trailers, rather than in commit metadata or on the forge [● Anthropic playbook, L1–L4, 2026](#).

**Compliance neither forces this archetype nor forbids the next.** The strongest human-review requirement in any published framework sits exactly at this archetype's upper boundary — SLSA's Source Track Level 4 requires changes to protected branches be agreed by *"two or more trusted persons"* — and the same document writes the machine exception into the requirement: an organization *"MAY choose to grant a Trusted Robot a perpetual exception"*. The line frameworks actually draw is identity and control of the automated actor, not human versus machine; NIST's SSDF likewise defines review as *"code review... and/or code analysis"* — alternatives, by the framework's own wording [● Verification: Gates, Aug 2026](#). So a team here should be clear-eyed that its safety belief is a chosen discipline: the regimes it answers to would mostly permit the next boundary, and the platform terms it operates under already assume the review it performs (*"You are responsible for reviewing, testing, and validating any Output before use"*) [● GitHub ToS §J.4, 2026](#).

## 08 The posture that fits

This fits a team that believes review is the load-bearing control and should stay human — and that an agent's work should come through the same front door as anyone else's.

The safety belief underneath: *a human reads every change before it lands*. The honest edge: that belief is now the best-measured erosion on the spectrum — under agent volume, reviewers wait longer, look less, and approve more. Holding this posture is not the default it feels like; it means engineering the review's protection, because the archetype's own output volume is the thing wearing it away.

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### — Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1 **STUDY** This project, ADR-0004 — the taxonomy contract: definitions, boundary criterion, the OSS-governance naming rationale. Adopted 2026-08-31.
- 2 **STUDY** This project, `research/practitioner-exemplars.md` — the large middle: ~57 exemplars with stake recorded; the constraint-moved-to-review pattern; Atlassian's eligibility classifier. Closed 2026-08-27.
- 3 **VENDOR** Stripe Engineering, "Minions: Stripe's one-shot, end-to-end coding agents" (Parts 1 and 2). Throughput figure with no control period, defect or revert rate. Stake: unaffiliated practitioner; mild recruiting incentive. 9 / 19 Feb 2026.
- 4 **VENDOR** Monzo, "Building Agent Chip". Explicitly a recruiting post. 13 Aug 2026.
- 5 **VENDOR** Dropbox, "Beyond code generation: rethinking engineering productivity in the age of AI agents" — the four-stage measurement model; metrics named, no values published. 28 May 2026.
- 6 **ANECDOTE** The Pragmatic Engineer, "How Uber uses AI for development" — secondary account of a Pragmatic Summit talk by Uber engineering leadership; paywalled past the preview, not circumvented. 10 Mar 2026.
- 7 **VENDOR** Stephen Toub (Microsoft), "Ten months with CCA in dotnet/runtime" — 878 agent PRs, 67.9% merged, 0.6% vs 0.8% revert; invited-only creation; the review-cost admission. Vendor on own product, but the data sits in a public repository and is independently checkable. 23 Mar 2026.
- 8 **VENDOR** GitHub Docs, "Risks and mitigations for GitHub Copilot cloud agent" — cannot approve or merge; additional human approval required. Accessed 2026-08-27.
- 9 **STUDY** Within-reviewer longitudinal study, arXiv:2606.22721 — 400 repeat reviewers, 11,429 reviews: agent-PR approval 30.1%→36.8%, inline comments -22%, latency +3.5×, human-PR approval declining over the same period. Jun 2026.

- 10 **STUDY** Duma et al., EASE 2026, arXiv:2605.02273 — mining the AIDev corpus (932,791 agentic PRs): most AI-generated PRs receive no reviews; review that occurs is automation-mediated. 4 May 2026.
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- 11 **VENDOR** LinearB, 2026 Software Engineering Benchmarks — 8.1M PRs, 4,800 teams: 5.3× reviewer pickup delay; acceptance 32.7% vs 84.5%; AI PRs ~154% larger. The AI/agentic PR classifier is not published. May 2026.
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- 12 **STUDY** This project, [research/tooling-landscape.md](#) — the merge boundary is platform-enforced; no product merges its own PR on its own judgement; the far end is reached by an administrator changing a policy. Closed 2026-08-30.
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- 13 **STUDY** This project, [research/verification-gates.md](#) — AI reviewers built unable to gate (Copilot Comment-only; Anthropic neutral check); the CodeRabbit 56.3%-rejection study; server-side-gate test; NIST PW.7 "and/or" and the SLSA robot exception, verified verbatim. Closed 2026-08-28.
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- 14 **STUDY** This project, [research/governance-legal-ip.md](#) — adoption vs authorship (USCO); attribution survival: platform-hosted, not repository-hosted. Closed 2026-08-30, corrected 2026-08-31.
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- 15 **VENDOR** GitHub Terms of Service §J.4 (contract text). Effective 2026-04-27.
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- 16 **VENDOR** This project, [research/ai-native-sdlc-playbook.md](#) — digest of Anthropic Academy's *The AI-Native SDLC Playbook* (14 lessons; © Anthropic, reference copy in [reference/](#)). The vendor's recommended practice for its own product: indicators defined for every play, values reported for none — nothing here is an outcome. Cited on this page: L1 (the shifts table — cite the plays, not the table), L2–L4 (the committed intent → spec → plan chain), L7 (reviewer attention), L9 (continuous evals), L10 (the PR review loop), L12 (the withheld production gate). Retrieved and digested 2026-09-02.
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